

Effectiveness of Crop Recommendation and Yield Prediction using Hybrid Moth Flame Optimization with Machine Learning

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ABSTRACT ORIGINAL

Agriculture is the main source of income, food, employment, and livelihood for most rural people in India. Several crops can be destroyed yearly due to a lack of technical skills and changing weather patterns such as rainfall, temperature, and other atmospheric parameters that play an enormous role in determining crop yield and profit. Therefore, selecting a suitable crop to increase crop yield is an essential aspect of improving real-life farming scenarios. Anticipating crop yield is one of the major concerns in agriculture and plays a critical role in global, regional, and field decision-making. Crop yield forecasting is based on crop parameters and meteorological, atmospheric, and soil conditions. This paper introduces a crop recommendation and yield prediction system using a Hybrid Moth Flame Optimization with Machine Learning (HMFO-ML) model. The presented HMFO-ML method effectively recommends crops and forecasts crop yield accurately and promptly. The proposed model used a Probabilistic Neural Network (PNN) for crop recommendation and the Extreme Learning Machine (ELM) method for the crop yield forecasting process. The HMFO algorithm was used to improve the forecasting rate of the ELM approach. A wide-ranging simulation analysis was carried out to evaluate the HMFO-ML model, showing its advantages over other models, as it exhibited a maximum R2 score of 98.82% and an accuracy of 99.67%. © by the authors.